BCSE429P

Cyber Physical Systems Design Lab

Laboratory 6 - DataFlow Models in PTolemy II

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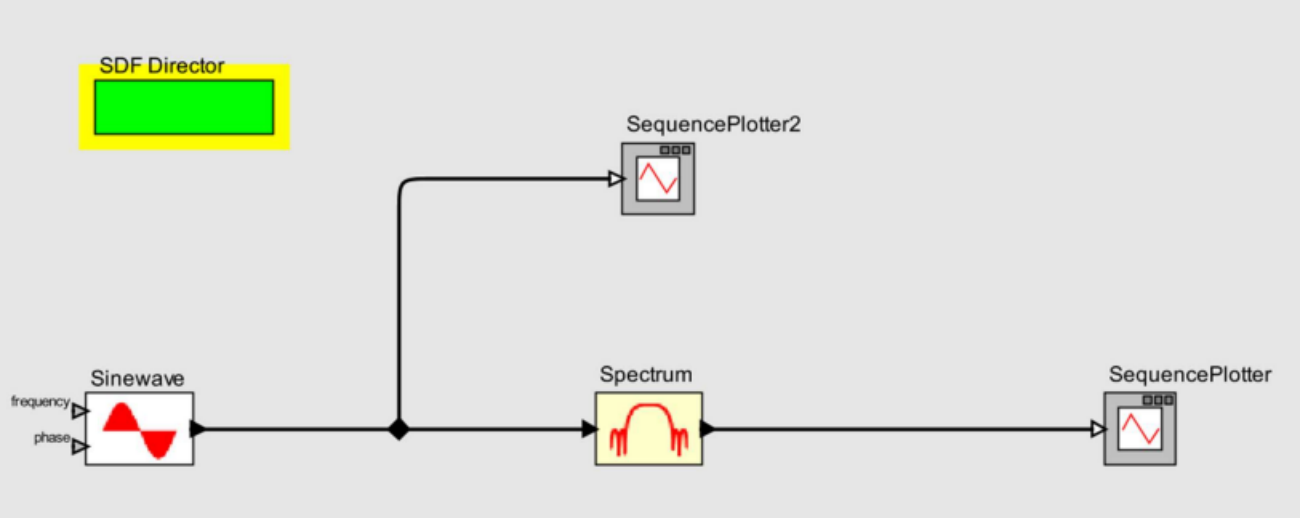
Reg. No.: 23BPS1011

Question:

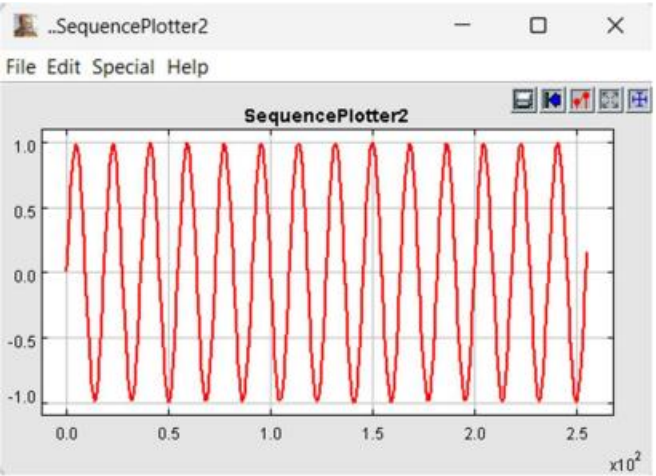
1. Practice the following problems from the Book Ptolemy II which is attached in the course page. From the book, look out the examples 3.1, 3.5 and 3.6, you can understand the blocks and plot the spectrum, entropy and normal sinusoidal waveform. In Figure 3,7, construct the output 0,2,4,6... 20 and 3,6,9... 21

2. Practice the following examples from the book given in this course: Figure 3.10, 3.11 and 3.13 for Dynamic Data Flow, Change the Iterations to 100 in each case (Wherever applicable). Figure 4.1, 4.2 and 4.3 - Process Networks - In all these exercises, replace the PN director with SDF/DDF and check their outputs.

Figure 3.1 – Sinusoidal Waveform and Spectrum



Normal Sinusoidal Waveform



Spectrum of Sinusoidal Signal

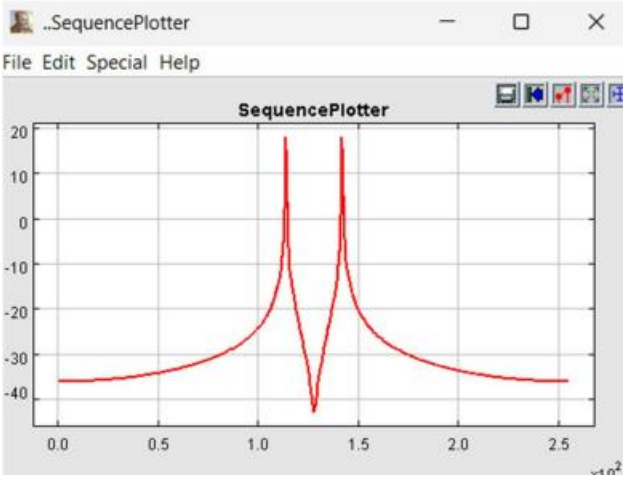


Figure 3.5 – Sound Spectrum

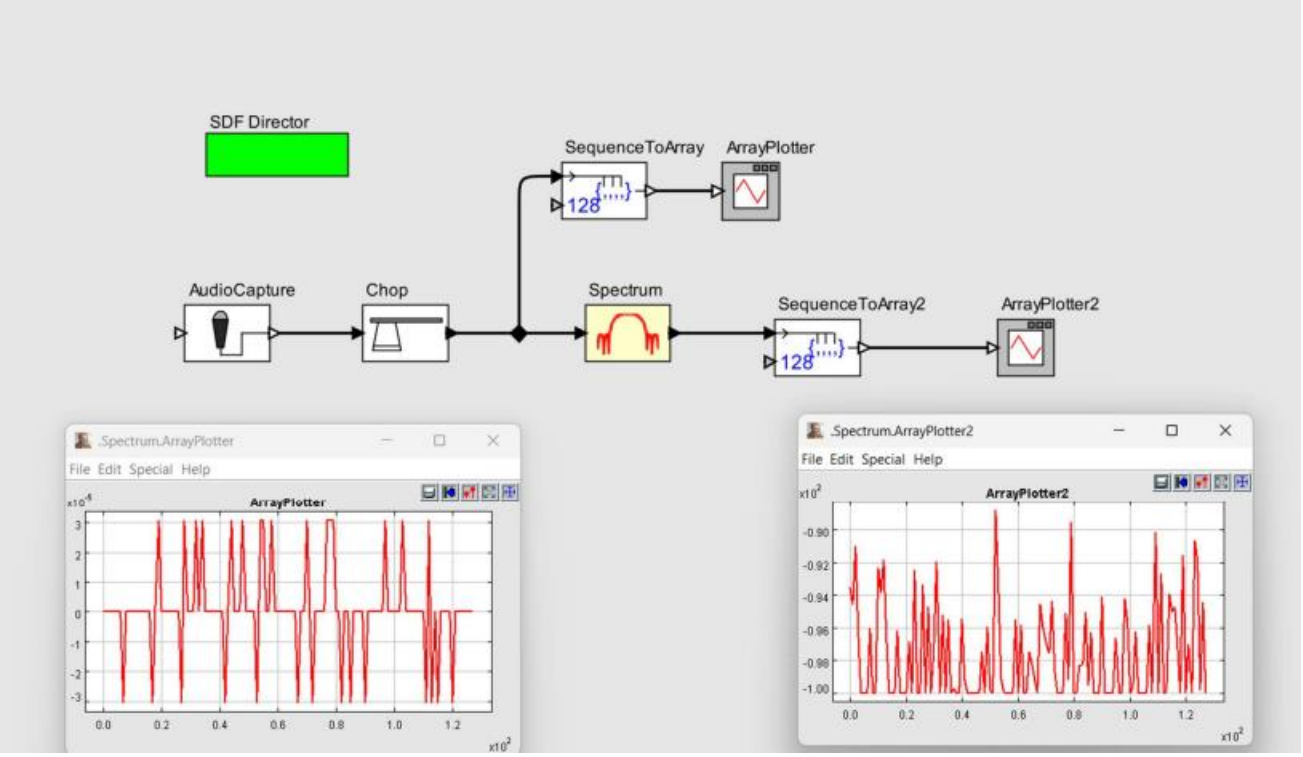


Figure 3.6 – Maximum Entropy Spectrum

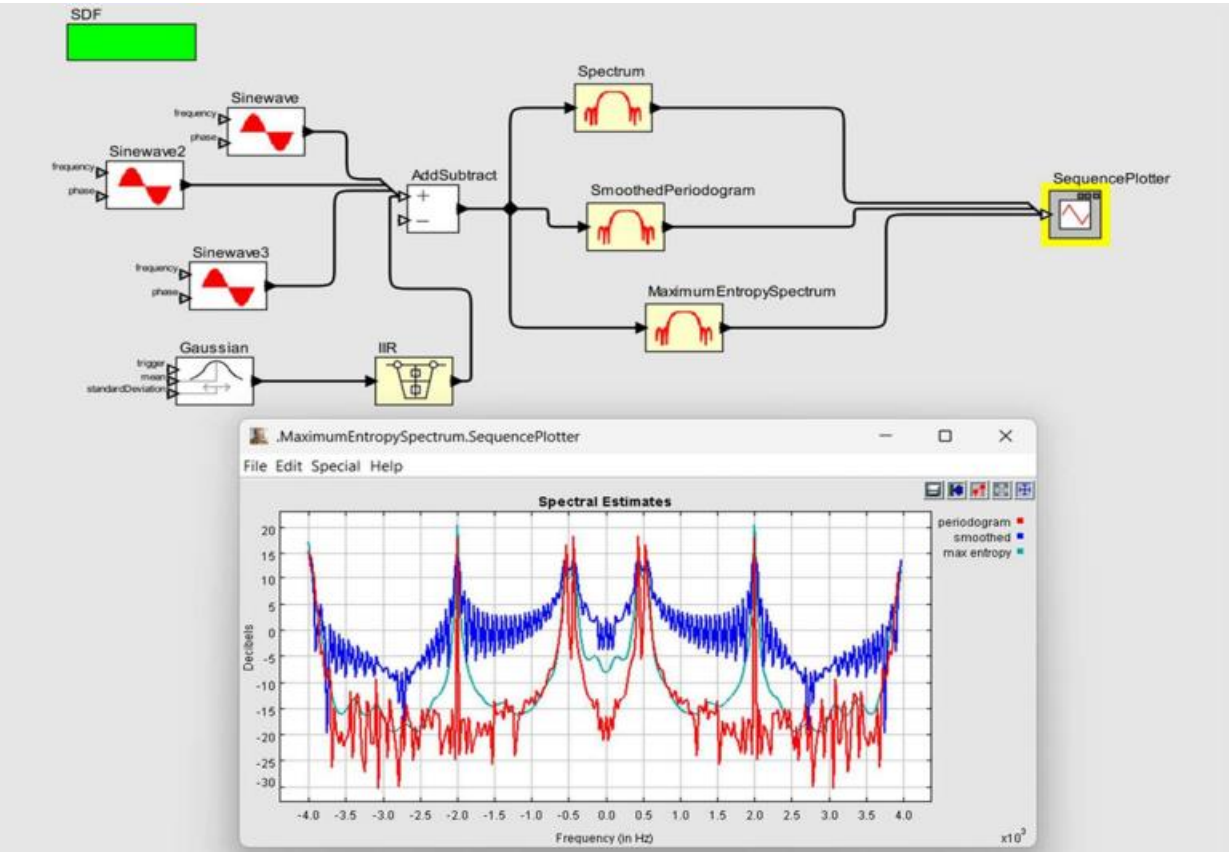
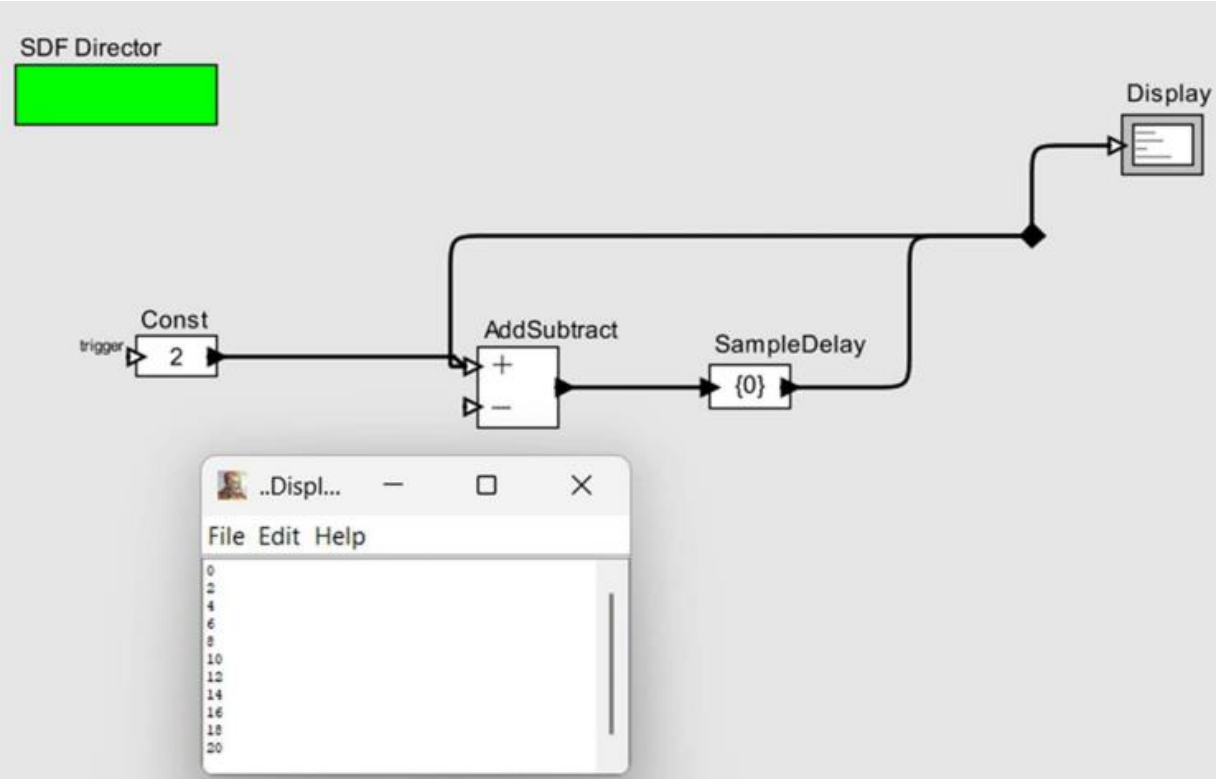


Figure 3.7 – Sequence Generation

Output 1



Output 2

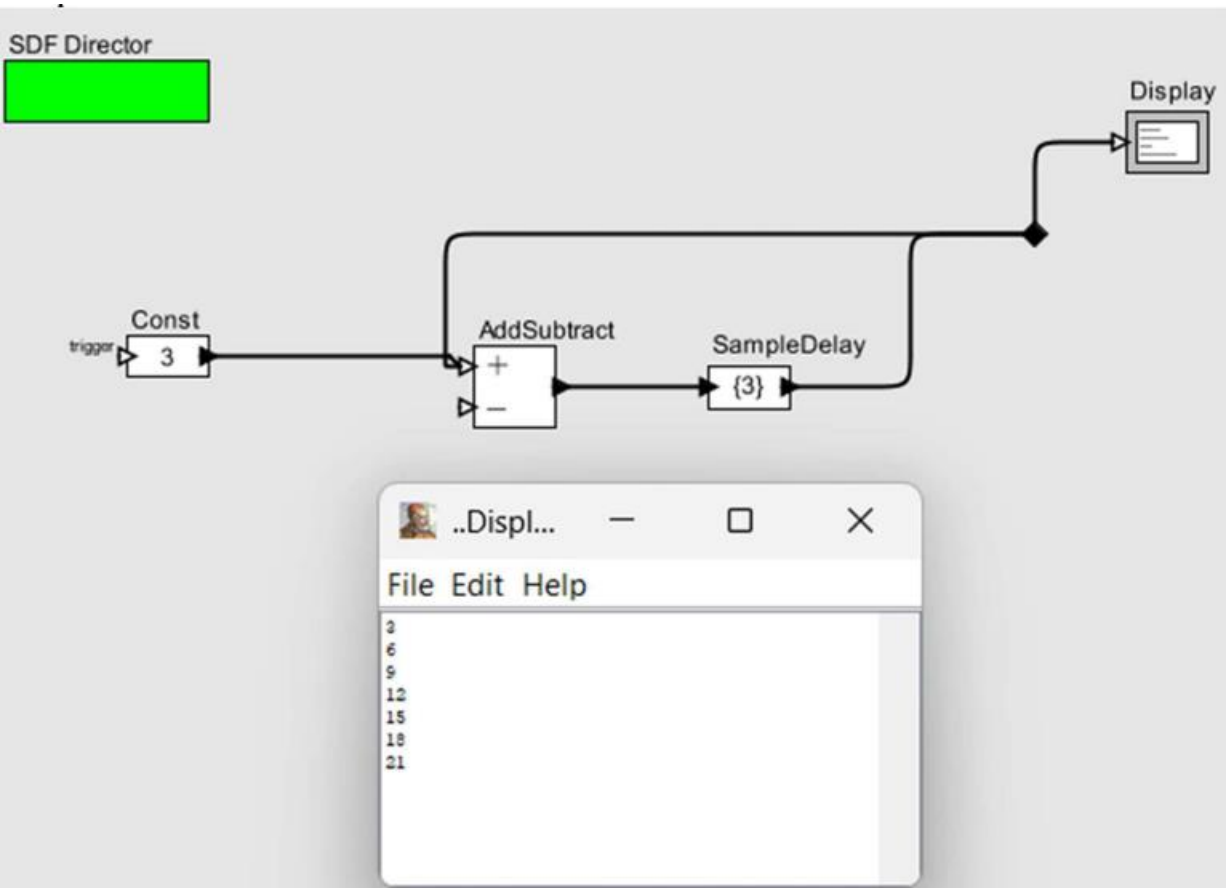
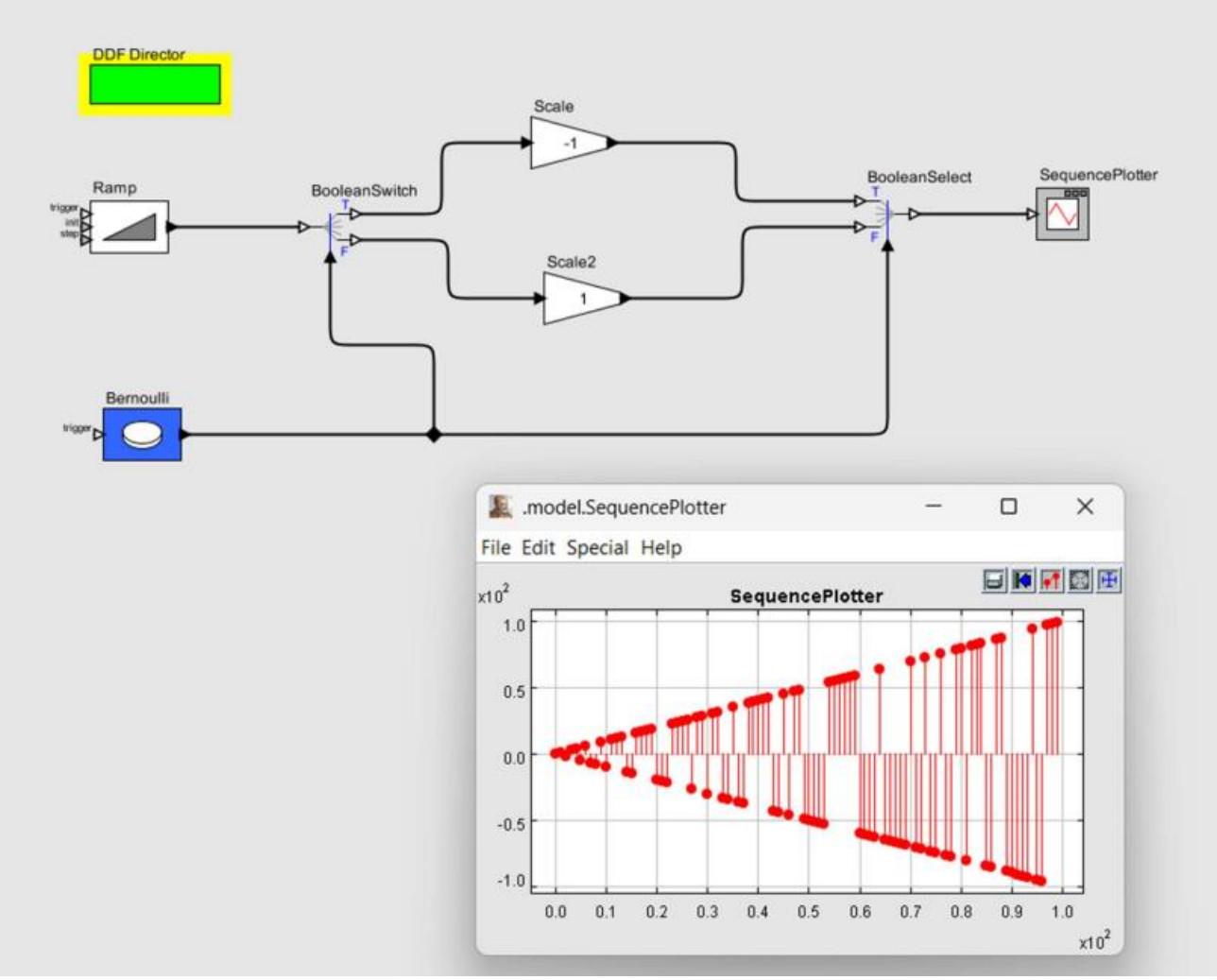


FIGURE 3.10 – Dynamic Data Flow



Note: DDF Director – Iterations = 100

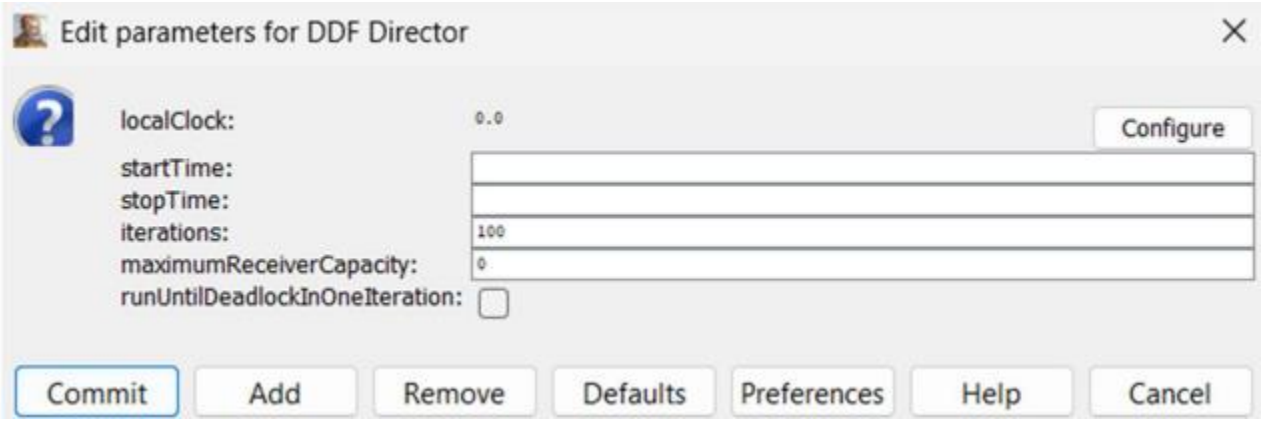
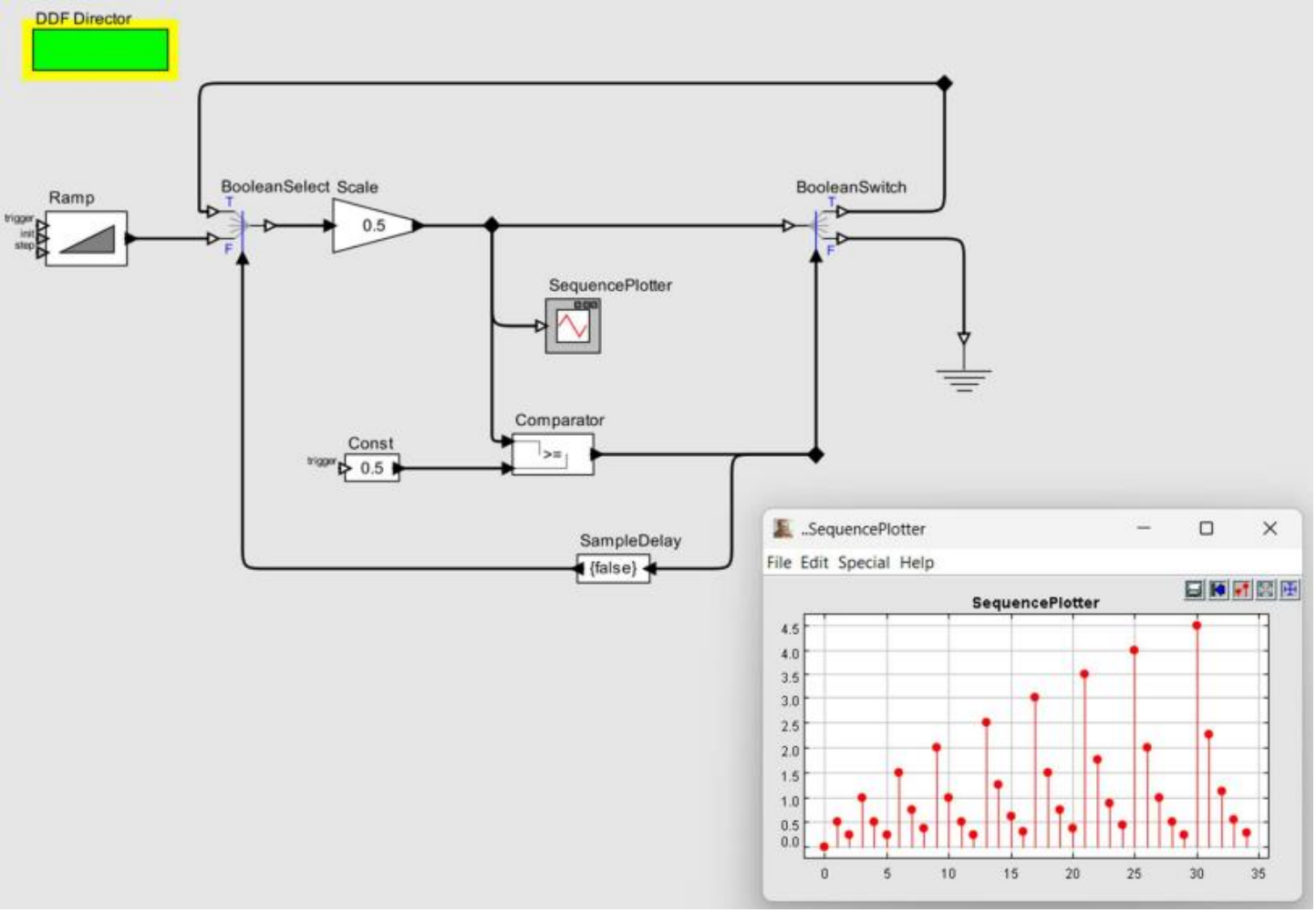


FIGURE 3.11 – Dynamic Data Flow



Note: DDF Director – Iterations = 100

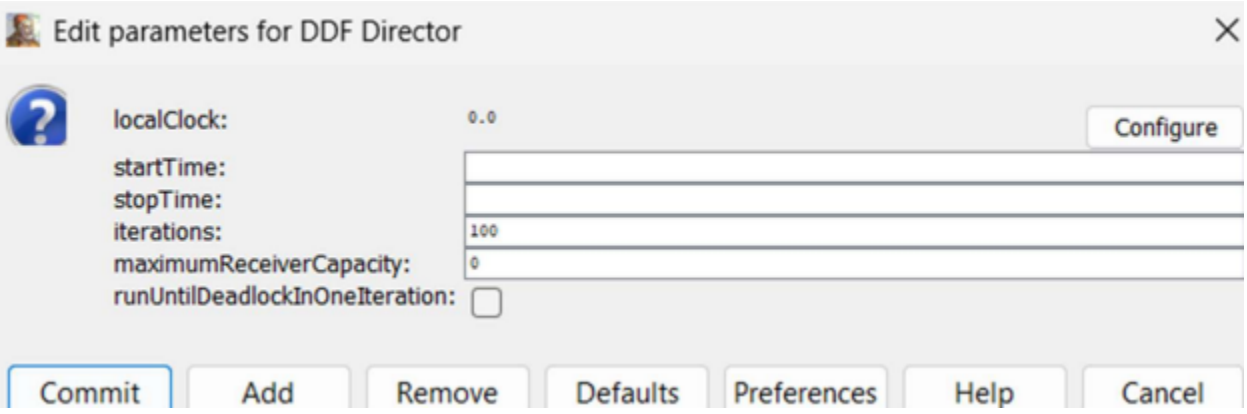
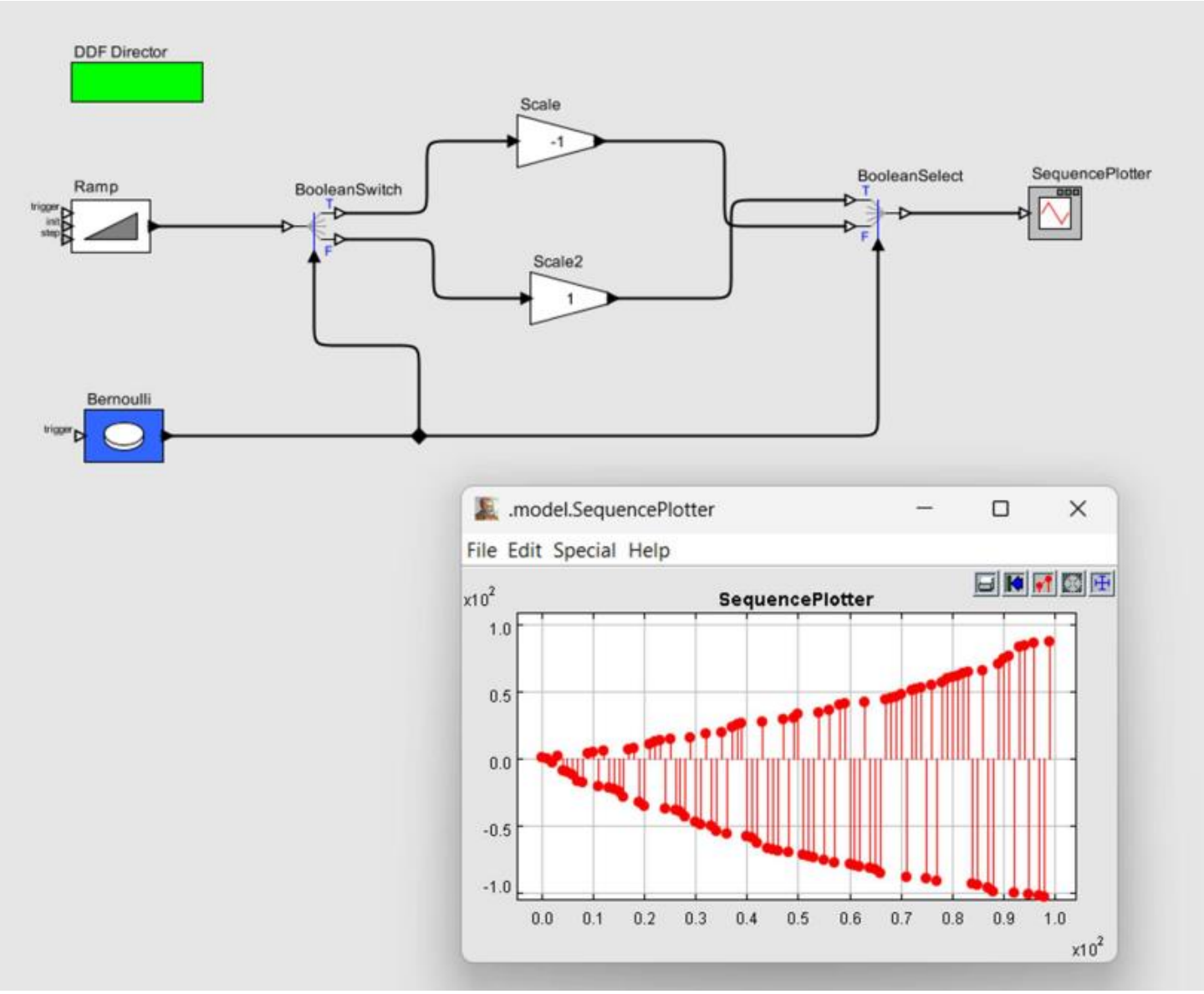


FIGURE 3.13 – Dynamic Data Flow



Note: DDF Director – Iterations = 100

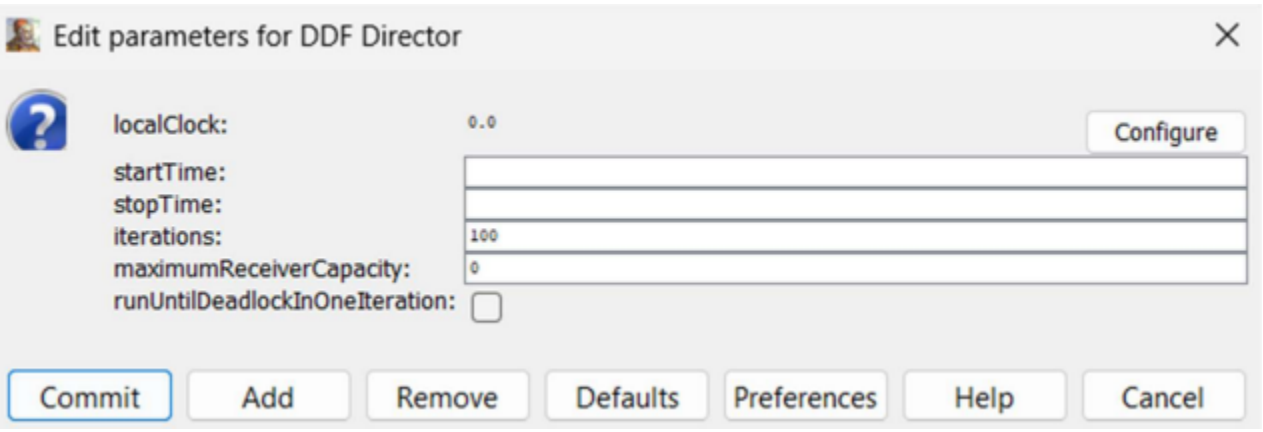
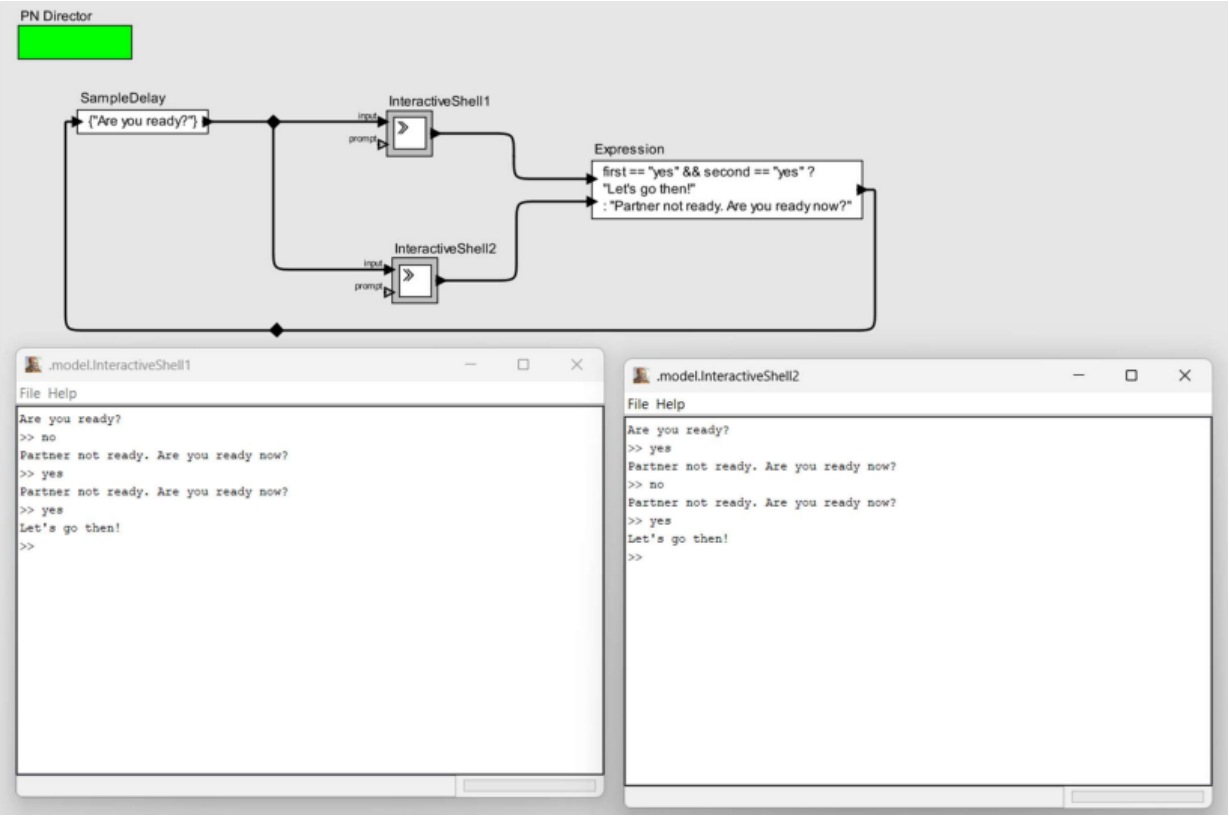
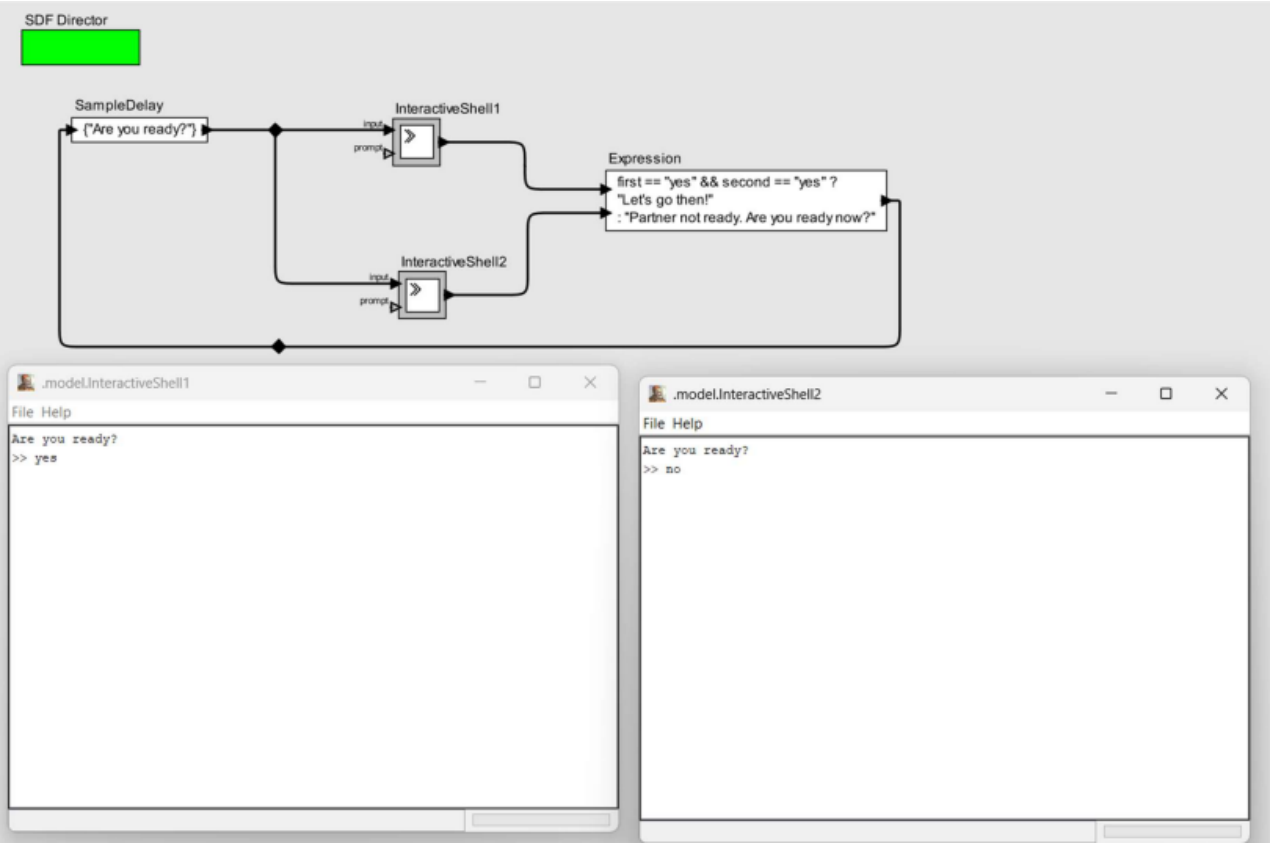


FIGURE 4.1 – Process Networks

PN Output



SDF Output



DDF Output

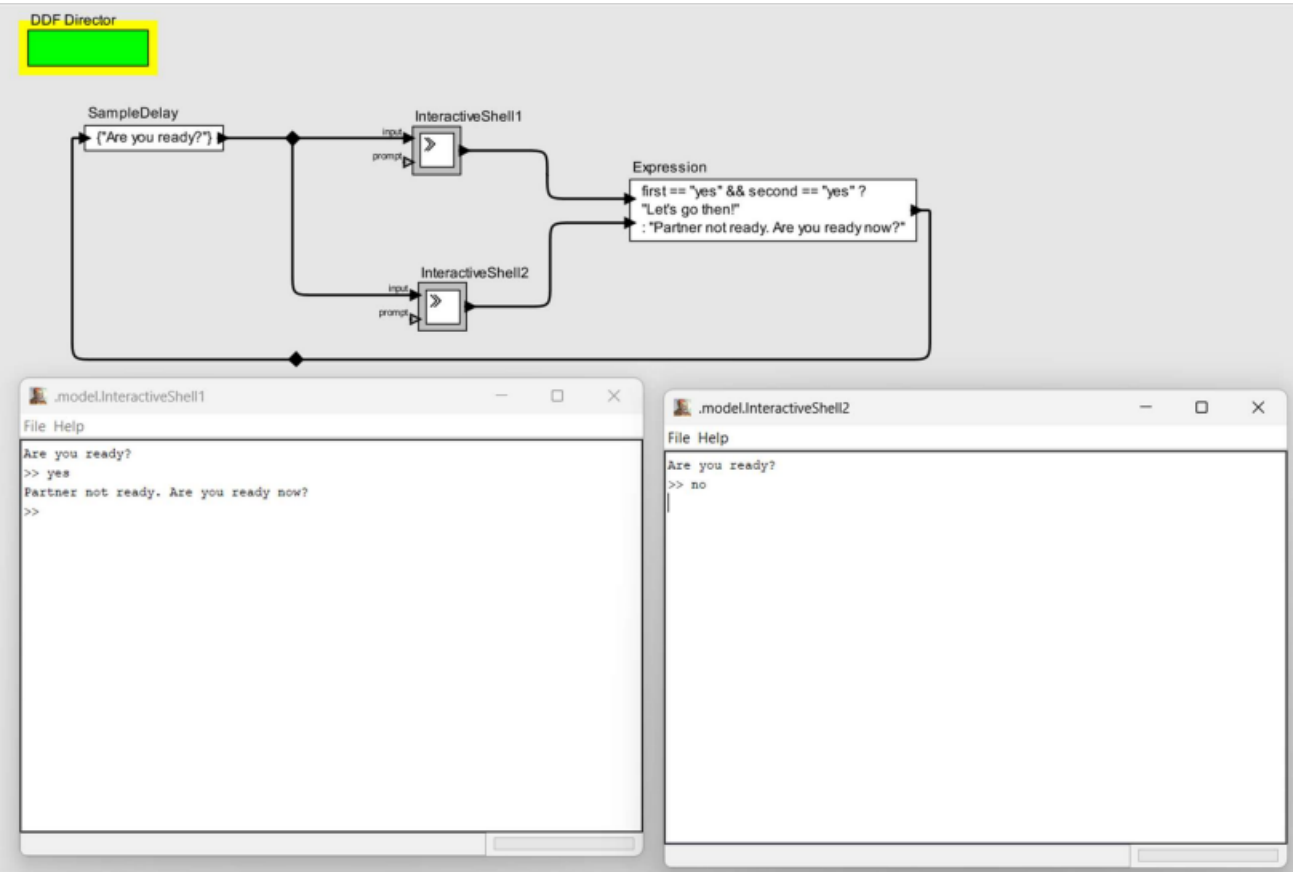
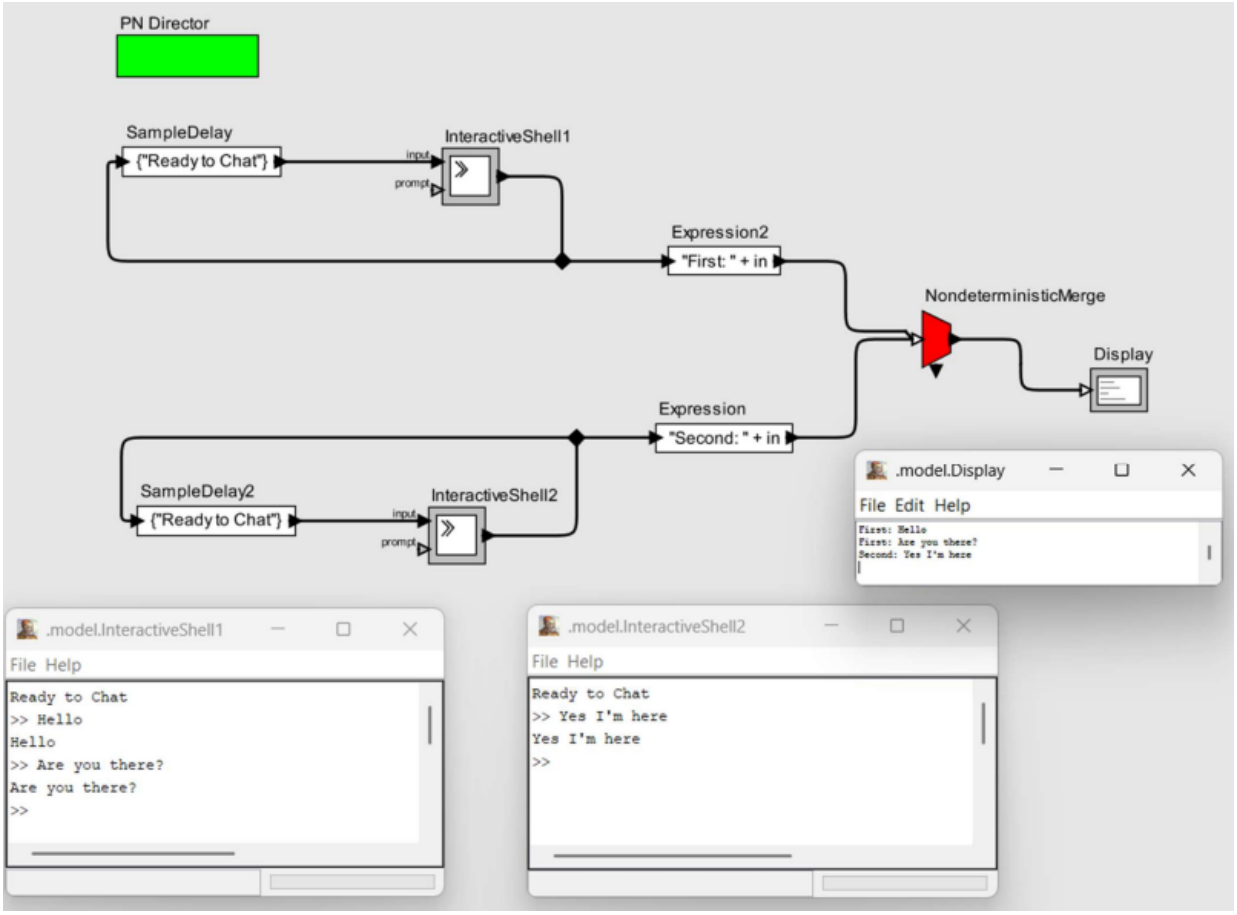
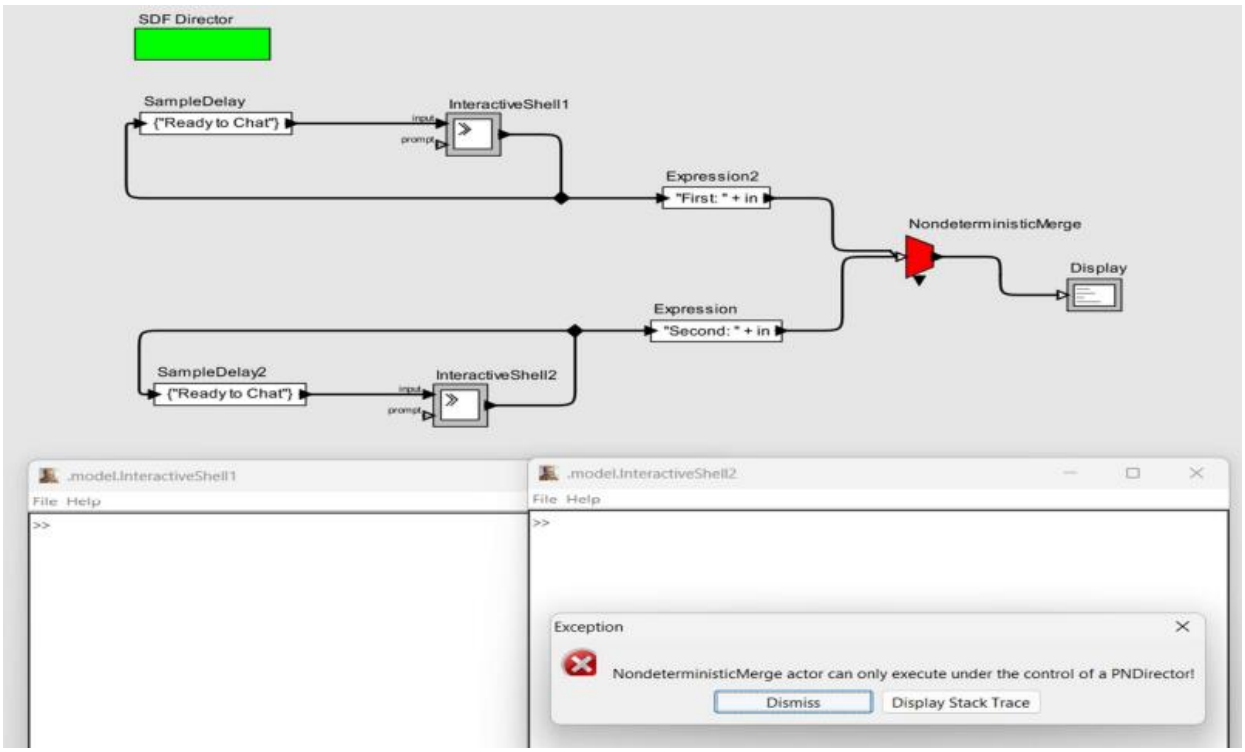


FIGURE 4.2 – Process Networks

PN Output:



SDF Output:



DDF Output:

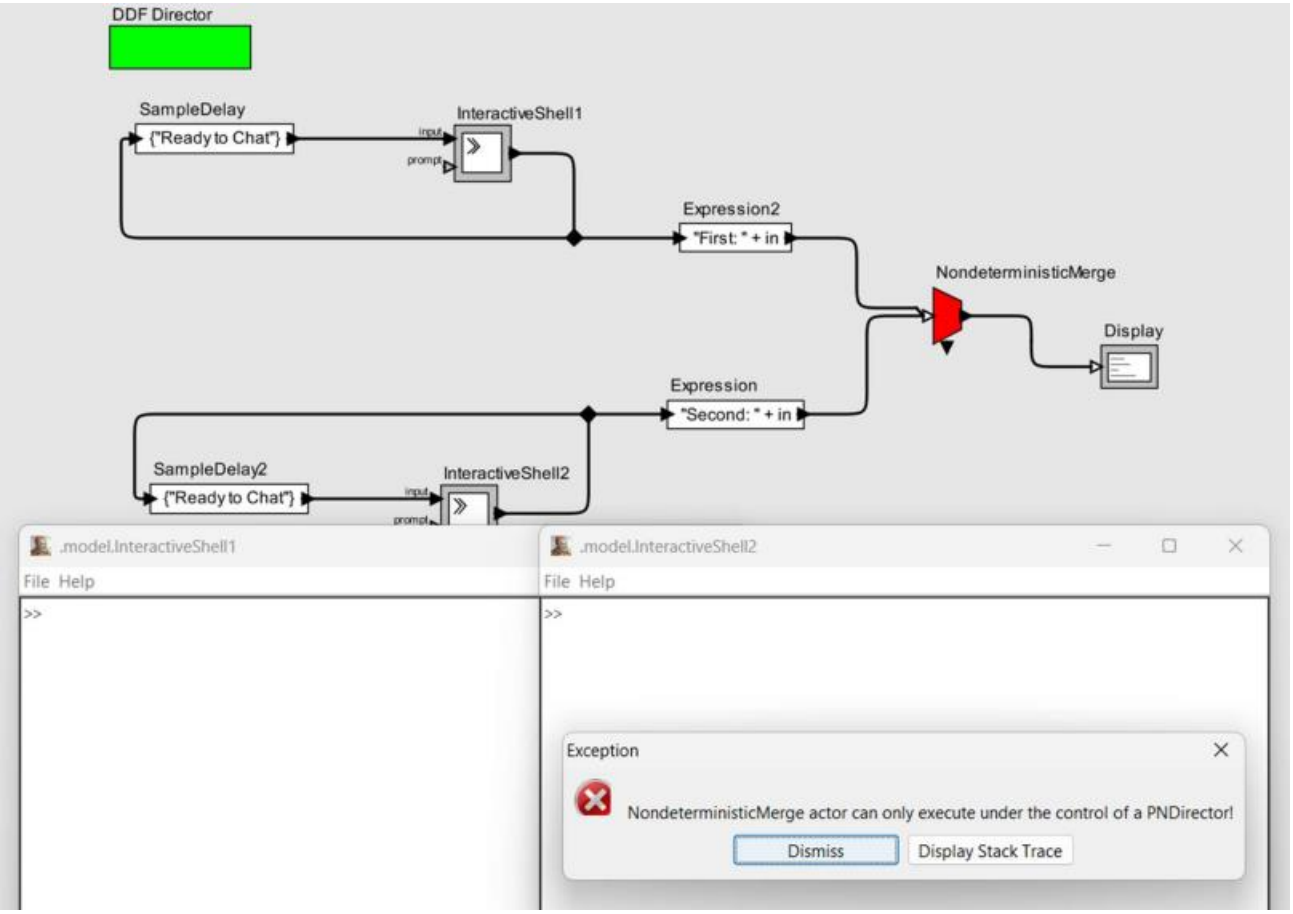
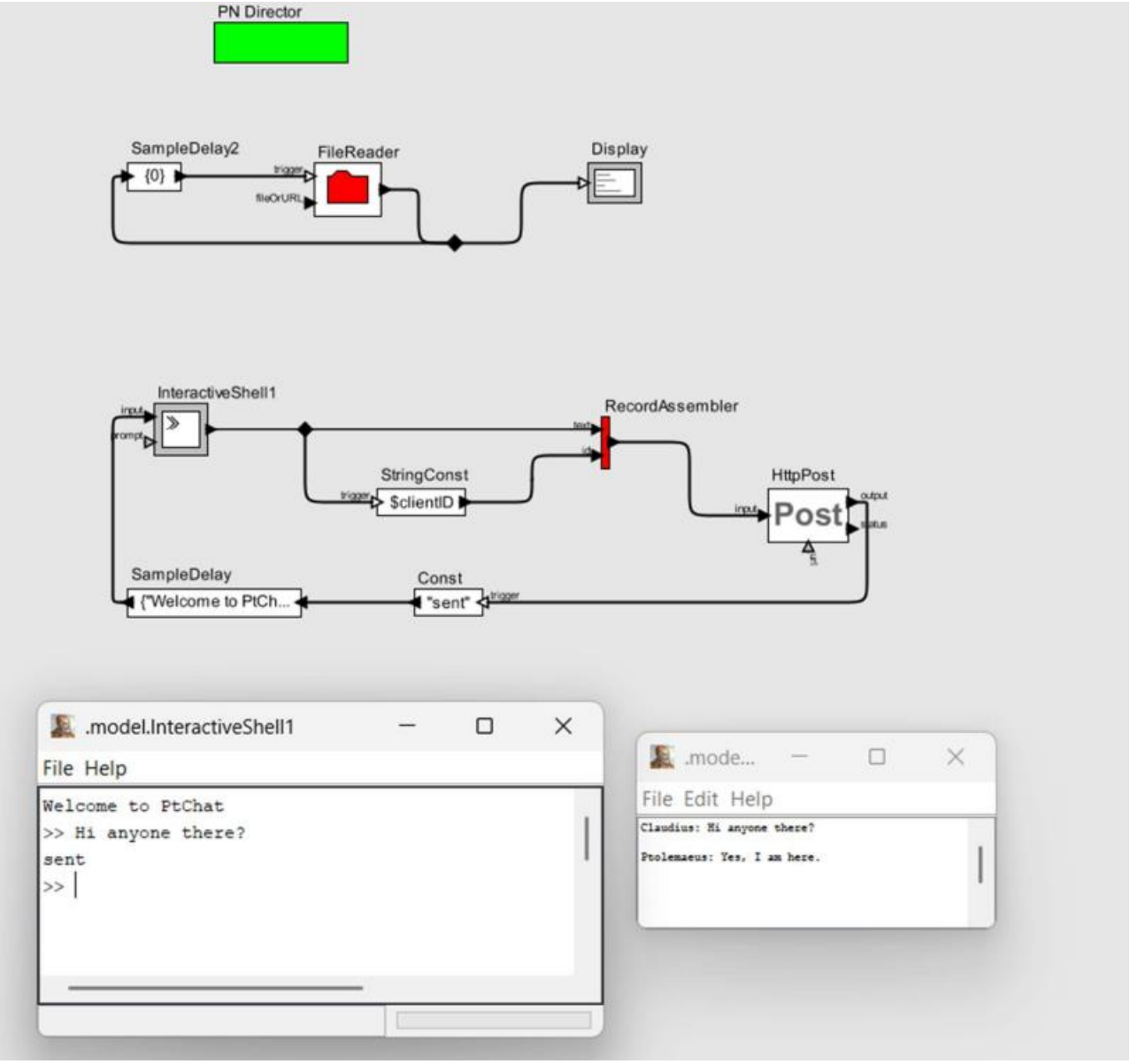
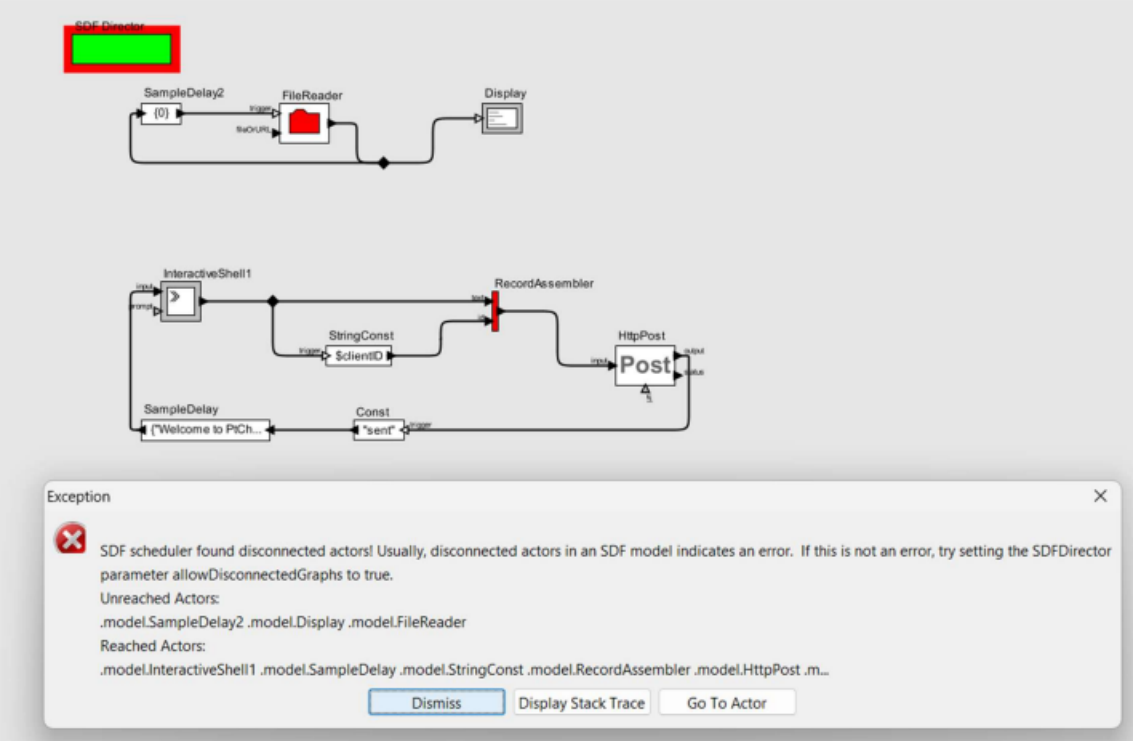


FIGURE 4.3 – Process Networks

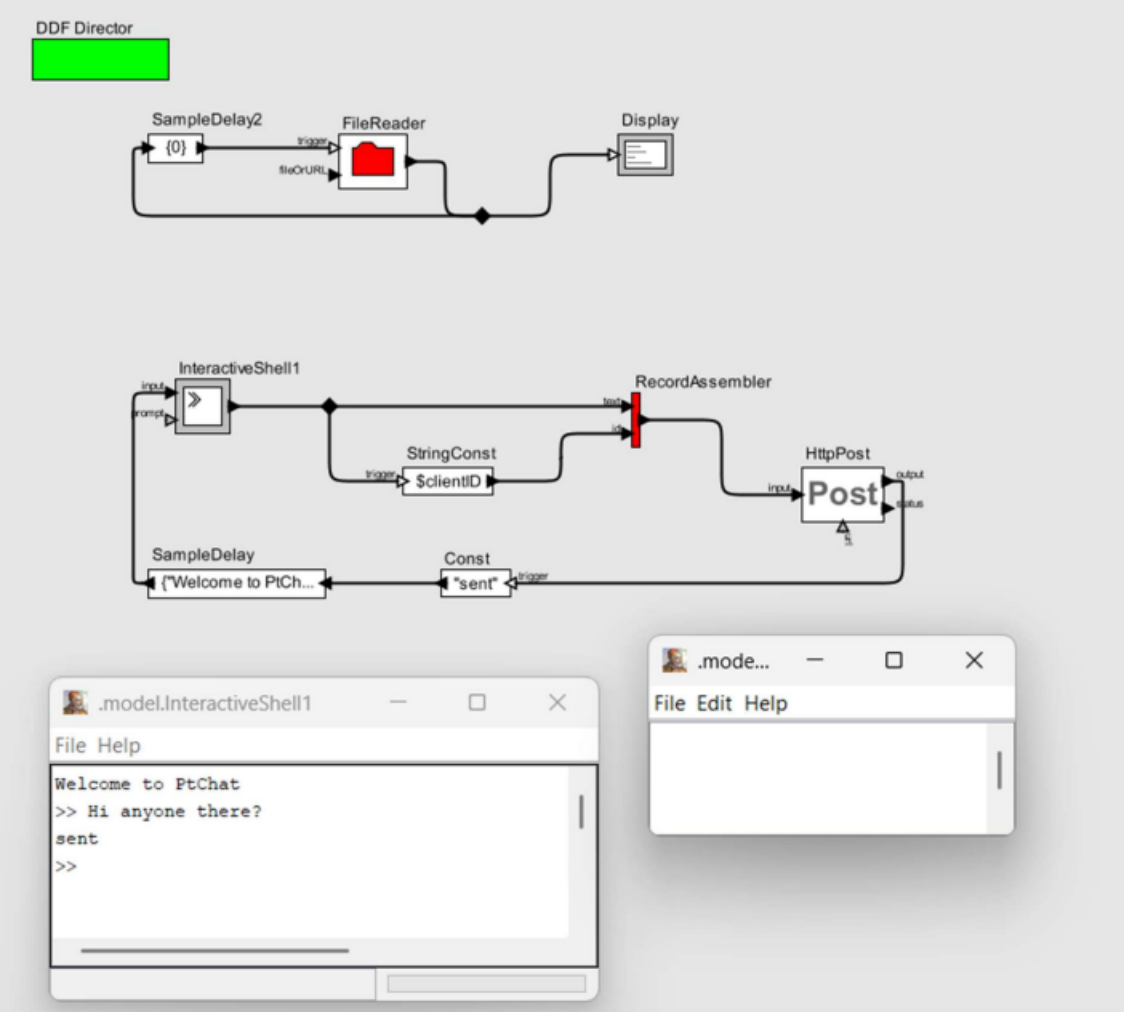
PN Output:



SDF Output:


SDF Director
SampleDelay2
{0}
FileReader
trigger
fileOrURL
Display
InteractiveShell1
input
prompt
RecordAssembler
StringConst
trigger
$clientID
HttpPost
input
Post
output
status
SampleDelay
{"Welcome to PtCh...
Const
"sent"
trigger
Exception
SDF scheduler found disconnected actors! Usually, disconnected actors in an SDF model indicates an error. If this is not an error, try setting the SDFDirector parameter allowDisconnectedGraphs to true.
Unreached Actors:
.model.SampleDelay2 .model.Display .model.FileReader
Reached Actors:
.model.InteractiveShell1 .model.SampleDelay .model.StringConst .model.RecordAssembler .model.HttpPost .m...
Dismiss
Display Stack Trace
Go To Actor

DDF Output:


DDF Director
SampleDelay2
{0}
FileReader
trigger
fileOrURL
Display
InteractiveShell1
input
prompt
RecordAssembler
StringConst
trigger
$clientID
HttpPost
input
Post
output
status
SampleDelay
{"Welcome to PtCh...
Const
"sent"
trigger
.model.InteractiveShell1
File Help
Welcome to PtChat
>> Hi anyone there?
sent
>>
.mode...
File Edit Help